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EXTENDING A MANAGEMENT FABRIC TO PERIPHERAL DEVICES FOR PERFORMANCE AND POWER OPTIMIZATION

Abstract

An information handling system includes a managed device coupled to a processor by a data communication interface. The management device manages a managed device via a management fabric different from the data communication interface.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] This disclosure generally relates to information handling systems, and more particularly relates to extending a management fabric to peripheral devices for performance and power optimization in an information handling system.

BACKGROUND

[0002] As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option is an information handling system. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes. Because technology and information handling needs and requirements may vary between different applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software resources that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

SUMMARY

[0003] An information handling system may include a managed device coupled to a processor by a data communication interface. The management device may manage a managed device via a management fabric different from the data communication interface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the Figures have not necessarily been drawn to scale. For example, the dimensions of some of the elements are exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the drawings presented herein, in which:

[0005] FIG. 1 is a block diagram illustrating an information handling system according to an embodiment of the present disclosure;

[0006] FIG. 2 is a flowchart illustrating a method for extending a management fabric to peripheral devices for performance and power optimization according to an embodiment of the present disclosure; and

[0007] FIG. 3 is a block diagram illustrating a generalized information handling system according to another embodiment of the present disclosure;

[0008] The use of the same reference symbols in different drawings indicates similar or identical items.

DETAILED DESCRIPTION OF DRAWINGS

[0009] The following description in combination with the Figures is provided to assist in understanding the teachings disclosed herein. The following discussion will focus on specific implementations and embodiments of the teachings. This focus is provided to assist in describing the teachings, and should not be interpreted as a limitation on the scope or applicability of the teachings. However, other teachings can certainly be used in this application. The teachings can also be used in other applications, and with several different types of architectures, such as distributed computing architectures, client/server architectures, or middleware server architectures and associated resources.

[0010] FIG. 1 illustrates hierarchical view of an information handling system **100** within a larger ecosystem that includes a cloud **140** and remote peripherals **142**. Information handling system **100** includes a hardware layer **110**, a pre-operating system (pre-OS) layer **120**, and an OS layer **130**. Hardware layer **110** includes a processor **112**, a baseboard management controller (BMC) **114**, and internal peripherals **116**. Processor **112** may be understood to represent one or more central

processing units (CPUs), controllers, or the like, and associated processing elements that operate to instantiate a host processing environment (that is, various elements of pre-OS layer **120** and OS layer **130**, as described below) on information handling system **100**. BMC **114** may represent one or more CPUs, controllers, or the like, and associated processing elements that operate to instantiate a management processing environment on information handling system **100** (that is, various elements of pre-OS layer **120** and OS layer **130**, as described below).

[0011] Internal peripherals **116** represent various hardware elements that are provided in information handling system **100**, typically within the confines of a case of the information handling system, that provide various functionalities or features that are not typically associated with the core processing functionality of processor **112**. Internal peripherals **116** may represent various hardware elements that are provided “down,” that is, fabricated on a printed circuit board (PCB) of information handling system **100**. For example, a hardware element that is down on a PCB of information handling system **100** may include integrated network interface controllers, audio controllers, video or graphics controllers, or the like. Internal peripherals **116** may further represent various hardware elements that can be installed into the PCB of information handling system **100** or removed from the PCB as needed or desired. For example, a hardware element that can be installed or removed from information handling system **100** may include stand-alone network interface cards (NICs), audio cards, video or graphics cards, or the like. Internal peripherals **116** are connected to processor **112** and BMC **114** via one or more data communication interfaces provided within hardware **110**.

[0012] Pre-OS level **120** includes Basic Input/Output System (BIOS)/Universal Extensible Firmware Interface (UEFI) firmware **122** and BMC firmware **124**. BIOS/UEFI **122** represents code operable to detect the elements of hardware **110**, and particularly of processor **112**, internal peripherals **116**, and remote peripherals **142**, to provide drivers initialize the elements, and to access the elements during run time operations. BMC firmware **124** represents code operable to detect the elements of hardware **110**, and particularly of BMC **114**, to provide drivers initialize the elements, and to access the elements during run time operations. The code associated with BIOS/UEFI **122** and with BMC firmware **124** is typically stored on non-volatile memory devices associated with processor **112** and BMC **114**, respectively. However, in operation pre-OS level **120** represents pre-OS operating environments instantiated on processor **112** for BIOS/UEFI **122**, and on BMC **114** for BMC firmware **124**.

[0013] OS level **130** includes a host OS **132**, a service OS **134**, and a BIOS/BMC interface **136**. Host OS **132** represents an operating system that is instantiated on processor **112** and that is launched by BIOS/UEFI **122** to provide an operating environment to perform the functions and features associated with the typical use of information handling system **100**. For example, host OS **132** may provide an environment that permits a user of information handling system **100** to launch programs and applications to perform desired functions, as needed or desired. In another example, host OS **132** may represent a commercial operating system, such as a Windows OS, a Linux OS, or the like, or may represent a dedicated operating system associated with a particular controller or usage model, as needed or desired. Service OS **134** represents an operating environment that is instantiated on BMC **114** to provide a user interface to the functions and features of the BMC through a network connected mechanism. In particular, service OS **134** permits a technician who is remote from information handling system **100** to access BMC **114**. BIOS/BMC interface **136** provides a bridge between the operating environment established on processor **112** and the operating environment established on BMC **114**, as described further below.

[0014] Cloud **140** broadly represents all elements of the ecosystem that reside outside the confines of information handling system **100**. More particularly, cloud **140** may represent networking resources, such as a local area network, an access network, an internet, the Internet, or the like, that permit information handling system **100** to communicate information outside the confines of the information handling system. Cloud **140** may further represent processing resources that may be

called upon by information handling system **100**, such as near-, mid-, or far-edge processing resources, or the like. Cloud **140** is typically accessed by information handling system **100** by a network connection, a wireless connection, a near-field connection, or the like. Cloud **140** may be accessed by information handling system **100** via one or more hardware element of hardware **100**. However the access to the hardware elements of hardware **100** will be mediated through one or more of the elements of pre-OS layer **120** and of OS layer **130**, as needed or desired.

[0015] Remote peripherals **142** represent various devices that are situated outside the confines of information handling system **100**, and that provide various functionalities or features that are not typically associated with the core processing functionality of processor **112**. Examples of remote peripherals **142** may include docks or other expansion devices, smart displays, monitors, or the like, camera/video devices, microphones, speakers, or other multimedia devices, mobile devices such as cell phones, tablet devices, or other mobile devices that may be included in a personal network associated with information handling system **100**, or other remotely connected devices that are situated outside the confines of information handling system **100**. Remote peripherals **142** may be directly connected to information handling system **100**, such as through a network connection, a wireless connection such as a WiFi connection, a Bluetooth connection, and Near Field Communication (NFC) connection, or the like, or the remote peripherals may be connected to the information handling system via cloud **140**, for example where the remote peripherals include a native network connection.

[0016] Information handling system **100** provides a host environment across hardware **110**, pre-OS layer **120**, and OS layer **130**. In particular, the host environment consists of processor **112**, internal peripherals **116**, BIOS/UEFI **122**, and host OS **132**. The host environment is illustrated by the light solid lines interconnecting the elements of the host environment. Information handling system **100** further provides a management environment **150** across hardware **110**, pre-OS layer **120**, and OS layer **130**. Management environment **150** represents an out-of-band control plane for monitoring, managing, and maintaining the operations of information handling system **100**. In particular, management environment **150** provides an anchor and root of trust for information handling system **100**, data collection, thermal, battery, and power supply control for the information handling system, an out-of-band mechanism to provide firmware updates, and the like.

[0017] As such, management environment **150** consists of BMC **114**, control plane elements of internal peripherals **116**, BMC firmware **124**, service OS **132**, and BIOS/UEFI interface **136**, as illustrated by the heavy solid lines interconnecting the elements of the management environment. BMC **114** operates to instantiate management environment **150** to monitor, manage, and maintain the elements of processor **112** and of internal peripherals **116** through various out-of-band data communication interfaces, such as an Inter-Integrated Circuit (I2C) interface, an Improved I2C (I3C) interface, a Serial Peripheral Interface (SPI) interface, an enhanced SPI (ePSI) interface, or other types of data communication interfaces. Further, through the action of BIOS/BMC interface **136**, management environment **150** is provided with hooks into the host environment to monitor, manage, and maintain various aspects of the host environment, as needed or desired.

[0018] It has been understood by the inventors of the current disclosure that, as consumer demand for a seamless computing ecosystem increases, management of the resources of the ecosystem must evolve past the management of the resources only of the information handling system, as described above with respect to the management environment of information handling system **100**. In particular, consumers are demanding broader integration of the intelligent device to peripheral management experience, discovery responsiveness, auto connectivity framework, multi-system performance per watt, multi device security across the entire ecosystem, and the like. In a particular embodiment, information handling system **100** operates to extend the control and data plane established by management environment **150** to elements outside the confines of the information handling system. In particular, BMC **114** instantiates a management fabric **152** that extends the functionality of management environment **150** to cloud **140** and to remote peripherals **142**. As

such, management fabric **152** represents a mechanism whereby BMC **114** can communicate with cloud **140** and with remote peripherals **142** to monitor, manage, and maintain the activities of cloud-based applications and features, and of the remote peripherals.

[0019] In particular, management fabric **152** may be incorporated into the code associated with BMC firmware **124**, with service OS **134**, or within other elements of information handling system **100**. Management fabric **152** provides a standardized interface with cloud **140** and with remote peripherals **142** to enable secure and verified communications for device management and optimization. Cloud-based applications within cloud **140**, and remote peripherals **142**, in order to become manageable via management fabric **152**, will incorporate a management fabric agent that complements the management fabric and permits the cloud-based applications and the remote peripherals to be managed by BMC **150**. The management fabric agent incorporates communication security modules that permit the root of trust for management activities to be rooted in BMC **114**. Once the communications under management fabric **152** are secured and rooted, the management fabric agent provides BMC **114** with access to management and optimization capabilities within the cloud-based applications and remote peripherals **142**. In accessing the management capabilities of the cloud-based applications and remote peripherals **142**, BMC **114** may manage device sleep- or power-states, firmware updates, custom device states, or the like. In accessing the optimization capabilities of the cloud-based applications and remote peripherals **142**, BMC **114** may optimize the operating state of the workloads and operating conditions to achieve a particular goal, such as low-power operations, high-performance operations, battery-longevity operations, low-acoustics operations, or the like.

[0020] In a particular embodiment, management fabric **152** represents a vendor specific protocol stack, including a physical interface layer that is provided on information handling system **100** and on remote peripherals **142**. In this way, a proprietary management capability is provided on compliant information handling systems and remote peripherals, as needed or desired. In another embodiment, management fabric **152** represents a physical interface agnostic protocol stack that can utilize various data communication interfaces to carry management fabric protocol information between information handling system **100**, cloud **140**, and remote peripherals **142**. For example, management fabric **152** may utilize a management-over-X (MoX) scheme, such as a management-over-Ethernet (MoE) scheme, or the like, where management fabric protocol information is encapsulated within data transactions native to another protocol, as needed or desired.

[0021] When information handling system **100** is in particular low-power states, such as a modern-standby state, BMC **114** remains powered and executing code to maintain management environment **150**. In this way, BMC **114** retains the ability to manage the low-power operations of internal peripherals **116**, powering the internal peripherals off and on, or otherwise suspending and resuming operations, as needed or desired to maintain a desired user experience. The existence of management fabric **152** provides the ability to extend this user experience management from within the confines of information handling system **100** to remote cloud **140** and to remote peripherals **142**. That is, BMC **114** may now suspend or resume applications within cloud **140**, and power off and on remote peripherals **142** to extend the user experience to the entire ecosystem.

[0022] FIG. 2 illustrates a method **200** for extending a management fabric to peripheral devices for performance and power optimization, starting at block **202**. A management device configured to establish a management fabric, as described above, operates to discover management fabric enabled devices within the ecosystem of an information handling system in block **204**. The detected fabric enabled devices are authenticated by the management device in block **206**. The management device initiates management sessions on the fabric enabled device in block **208**, and the method ends in block **210**. While a management fabric was described in the context of an information handling system with a BMC, as described above, this is not necessarily so.

[0023] For example, a management device may include a BMC, a management and optimization manager, or other element of an information handling system configured to monitor, manage,

maintain, and optimize the operations of the information handling system and its associated peripheral devices, as needed or desired. Such a management device may operate in band with a host environment of the information handling system, or out of band from the host environment, as described above. In a particular case, the management fabric may be established by a separate optimization environment instantiated on an information handling system, as needed or desired. Note further that the management fabric agent was described as being associated with cloud-based applications or remote peripherals, as described above, but this is not necessarily the only cases for the implementation of a management fabric agent. For example, internal peripherals of an information handling system may likewise benefit from the inclusion of a management fabric agent, thereby providing a common framework for management of a wider range of devices, within the confines of the information handling system, or remote from the information handling system, as needed or desired.

[0024] As used herein a fabric may refer to a network topology, that is a set of physical data communication interface connections, that connects devices together. The network topology may include the supporting infrastructure needed to establish the connections, such as switches, routers, hubs, WiFi access points, and the like. A fabric may further refer to an abstracted overlay on the network topology that provides one or more separate virtualized data communication planes. The separate communication planes may each implement a particular protocol stack, and provide plane-dependent services and policies for data transfers on the communication planes. The data associated with a particular data plane can be encapsulated with plane-specific headers that identify the data as being associated with the particular data plane. Such fabric-dependent data packets can then be further encapsulated with other headers associated with the particular network topology, thereby making the data packets fabric-agnostic with respect to the physical network topology.

[0025] FIG. 3 illustrates a generalized embodiment of an information handling system **300** similar to information handling system **300**. For purpose of this disclosure an information handling system can include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, information handling system **300** can be a personal computer, a laptop computer, a smart phone, a tablet device or other consumer electronic device, a network server, a network storage device, a switch router or other network communication device, or any other suitable device and may vary in size, shape, performance, functionality, and price. Further, information handling system **300** can include processing resources for executing machine-executable code, such as a central processing unit (CPU), a programmable logic array (PLA), an embedded device such as a System-on-a-Chip (SoC), or other control logic hardware. Information handling system **300** can also include one or more computer-readable medium for storing machine-executable code, such as software or data. Additional components of information handling system **300** can include one or more storage devices that can store machine-executable code, one or more communications ports for communicating with external devices, and various input and output (I/O) devices, such as a keyboard, a mouse, and a video display. Information handling system **300** can also include one or more buses operable to transmit information between the various hardware components.

[0026] Information handling system **300** can include devices or modules that embody one or more of the devices or modules described below, and operates to perform one or more of the methods described below. Information handling system **300** includes a processors **302** and **304**, an input/output (I/O) interface **310**, memories **320** and **325**, a graphics interface **330**, a basic input and output system/universal extensible firmware interface (BIOS/UEFI) module **340**, a disk controller **350**, a hard disk drive (HDD) **354**, an optical disk drive (ODD) **356**, a disk emulator **360** connected to an external solid state drive (SSD) **362**, an I/O bridge **370**, one or more add-on resources **374**, a trusted platform module (TPM) **376**, a network interface **380**, a management device **390**, and a

power supply 395. Processors 302 and 304, I/O interface 310, memory 320, graphics interface 330, BIOS/UEFI module 340, disk controller 350, HDD 354, ODD 356, disk emulator 360, SSD 362, I/O bridge 370, add-on resources 374, TPM 376, and network interface 380 operate together to provide a host environment of information handling system 300 that operates to provide the data processing functionality of the information handling system. The host environment operates to execute machine-executable code, including platform BIOS/UEFI code, device firmware, operating system code, applications, programs, and the like, to perform the data processing tasks associated with information handling system 300.

[0027] In the host environment, processor 302 is connected to I/O interface 310 via processor interface 306, and processor 304 is connected to the I/O interface via processor interface 308. Memory 320 is connected to processor 302 via a memory interface 322. Memory 325 is connected to processor 304 via a memory interface 327. Graphics interface 330 is connected to I/O interface 310 via a graphics interface 332, and provides a video display output 336 to a video display 334. In a particular embodiment, information handling system 300 includes separate memories that are dedicated to each of processors 302 and 304 via separate memory interfaces. An example of memories 320 and 330 include random access memory (RAM) such as static RAM (SRAM), dynamic RAM (DRAM), non-volatile RAM (NV-RAM), or the like, read only memory (ROM), another type of memory, or a combination thereof.

[0028] BIOS/UEFI module 340, disk controller 350, and I/O bridge 370 are connected to I/O interface 310 via an I/O channel 312. An example of I/O channel 312 includes a Peripheral Component Interconnect (PCI) interface, a PCI-Extended (PCI-X) interface, a high-speed PCI-Express (PCIe) interface, another industry standard or proprietary communication interface, or a combination thereof. I/O interface 310 can also include one or more other I/O interfaces, including an Industry Standard Architecture (ISA) interface, a Small Computer Serial Interface (SCSI) interface, an Inter-Integrated Circuit (I2C) interface, a System Packet Interface (SPI), a Universal Serial Bus (USB), another interface, or a combination thereof. BIOS/UEFI module 340 includes BIOS/UEFI code operable to detect resources within information handling system 300, to provide drivers for the resources, initialize the resources, and access the resources. BIOS/UEFI module 340 includes code that operates to detect resources within information handling system 300, to provide drivers for the resources, to initialize the resources, and to access the resources.

[0029] Disk controller 350 includes a disk interface 352 that connects the disk controller to HDD 354, to ODD 356, and to disk emulator 360. An example of disk interface 352 includes an Integrated Drive Electronics (IDE) interface, an Advanced Technology Attachment (ATA) such as a parallel ATA (PATA) interface or a serial ATA (SATA) interface, a SCSI interface, a USB interface, a proprietary interface, or a combination thereof. Disk emulator 360 permits SSD 364 to be connected to information handling system 300 via an external interface 362. An example of external interface 362 includes a USB interface, an IEEE 1394 (Firewire) interface, a proprietary interface, or a combination thereof. Alternatively, solid-state drive 364 can be disposed within information handling system 300.

[0030] I/O bridge 370 includes a peripheral interface 372 that connects the I/O bridge to add-on resource 374, to TPM 376, and to network interface 380. Peripheral interface 372 can be the same type of interface as I/O channel 312, or can be a different type of interface. As such, I/O bridge 370 extends the capacity of I/O channel 312 where peripheral interface 372 and the I/O channel are of the same type, and the I/O bridge translates information from a format suitable to the I/O channel to a format suitable to the peripheral channel 372 where they are of a different type. Add-on resource 374 can include a data storage system, an additional graphics interface, a network interface card (NIC), a sound/video processing card, another add-on resource, or a combination thereof. Add-on resource 374 can be on a main circuit board, on separate circuit board or add-in card disposed within information handling system 300, a device that is external to the information handling system, or a combination thereof.

[0031] Network interface **380** represents a NIC disposed within information handling system **300**, on a main circuit board of the information handling system, integrated onto another component such as I/O interface **310**, in another suitable location, or a combination thereof. Network interface device **380** includes network channels **382** and **384** that provide interfaces to devices that are external to information handling system **300**. In a particular embodiment, network channels **382** and **384** are of a different type than peripheral channel **372** and network interface **380** translates information from a format suitable to the peripheral channel to a format suitable to external devices. An example of network channels **382** and **384** includes InfiniBand channels, Fibre Channel channels, Gigabit Ethernet channels, proprietary channel architectures, or a combination thereof. Network channels **382** and **384** can be connected to external network resources (not illustrated). The network resource can include another information handling system, a data storage system, another network, a grid management system, another suitable resource, or a combination thereof.

[0032] Management device **390** represents one or more processing devices, such as a dedicated baseboard management controller (BMC) System-on-a-Chip (SoC) device, one or more associated memory devices, one or more network interface devices, a complex programmable logic device (CPLD), and the like, that operate together to provide the management environment for information handling system **300**. In particular, management device **390** is connected to various components of the host environment via various internal communication interfaces, such as a Low Pin Count (LPC) interface, an Inter-Integrated-Circuit (I2C) interface, a PCIe interface, or the like, to provide an out-of-band (OOB) mechanism to retrieve information related to the operation of the host environment, to provide BIOS/UEFI or system firmware updates, to manage non-processing components of information handling system **300**, such as system cooling fans and power supplies. Management device **390** can include a network connection to an external management system, and the management device can communicate with the management system to report status information for information handling system **300**, to receive BIOS/UEFI or system firmware updates, or to perform other task for managing and controlling the operation of information handling system **300**. Management device **390** can operate off of a separate power plane from the components of the host environment so that the management device receives power to manage information handling system **300** where the information handling system is otherwise shut down. An example of management device **390** include a commercially available BMC product or other device that operates in accordance with an Intelligent Platform Management Initiative (IPMI) specification, a Web Services Management (WSMan) interface, a Redfish Application Programming Interface (API), another Distributed Management Task Force (DMTF), or other management standard, and can include an Integrated Dell Remote Access Controller (iDRAC), an Embedded Controller (EC), or the like. Management device **390** may further include associated memory devices, logic devices, security devices, or the like, as needed or desired.

[0033] Although only a few exemplary embodiments have been described in detail herein, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the embodiments of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the embodiments of the present disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures.

[0034] The above-disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover any and all such modifications, enhancements, and other embodiments that fall within the scope of the present invention. Thus, to the maximum extent allowed by law, the scope of the present invention is to be determined by the broadest permissible

interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

Claims

1. An information handling system, comprising: a processor; a management device; and a managed device, wherein the managed device is coupled to the processor by a data communication interface, and wherein the management device manages the managed device via a management fabric different from the data communication interface.
2. The information handling system of claim 1, wherein, in managing the managed device, the management device is configured to authenticate the managed device via the management fabric.
3. The information handling system of claim 2, wherein, in managing the managed device, the management device is further configured to manage at least one of a device sleep-state, a device power-state, a firmware update, and a custom device state on the managed device via the management fabric.
4. The information handling system of claim 3, wherein in managing the managed device, the management device is further configured to perform an optimization operation on the managed device via the management fabric.
5. The information handling system of claim 4, wherein in performing the optimization operation, the management device is further configured to optimize the managed device for at least one of low-power operation, high-performance operation, battery-longevity operation, and low-acoustics operation.
6. The information handling system of claim 1, wherein the managed device is an internal peripheral device of the information handling system.
7. The information handling system of claim 1, wherein the managed device is a remote peripheral device of the information handling system.
8. The information handling system of claim 1, wherein the management device is instantiated by the processor.
9. The information handling system of claim 1, wherein the management device operates out of band from the processor.
10. The information handling system of claim 9, wherein the management device includes a baseboard management controller.
11. A method, comprising: providing, in an information handling system, a processor; providing, in the information handling system, a management device; coupling a managed device to the processor by a data communication interface; and managing, by the management device, the managed device via a management fabric different from the data communication interface.
12. The method of claim 11 wherein, in managing the managed device, the method further comprises authenticating, by the management device, the managed device via the management fabric.
13. The method of claim 12 wherein, in managing the managed device, the method further comprises managing, by the management device, at least one of a device sleep-state, a device power-state, a firmware update, and a custom device state on the managed device via the management fabric.
14. The method of claim 13 wherein in managing the managed device, the method further comprises performing, by the management device, an optimization operation on the managed device via the management fabric.
15. The method of claim 14 wherein in performing the optimization operation, the method further comprises optimizing the managed device for at least one of low-power operation, high-performance operation, battery-longevity operation, and low-acoustics operation.
16. The method of claim 11, wherein the managed device is an internal peripheral device of the

information handling system.

17. The method of claim 11, wherein the managed device is a remote peripheral device of the information handling system.

18. The method of claim 11, wherein the management device is instantiated by the processor.

19. The method of claim 11, wherein the management device operates out of band from the processor.

20. An information handling system, comprising: a processor; a managed device coupled to the processor by a data communication interface; and a management device configured to instantiate a management fabric on the data communication interface, and to manage the managed device via the management fabric.
